Section-4

Inverse functions (or) Transformation

18. Inversion functions

Inverse of a fn
↳ Reverses the effect of the original fn.

Set X $\xrightarrow{\text{we have a fn. which maps}}$ Set Y

$\boxed{X \to Y}$

f^{-1} (inverse fn.) → maps y back to x

Given

a fn. $\boxed{f: X \to Y}$

inverse fn. $\boxed{f^{-1}: Y \to X}$

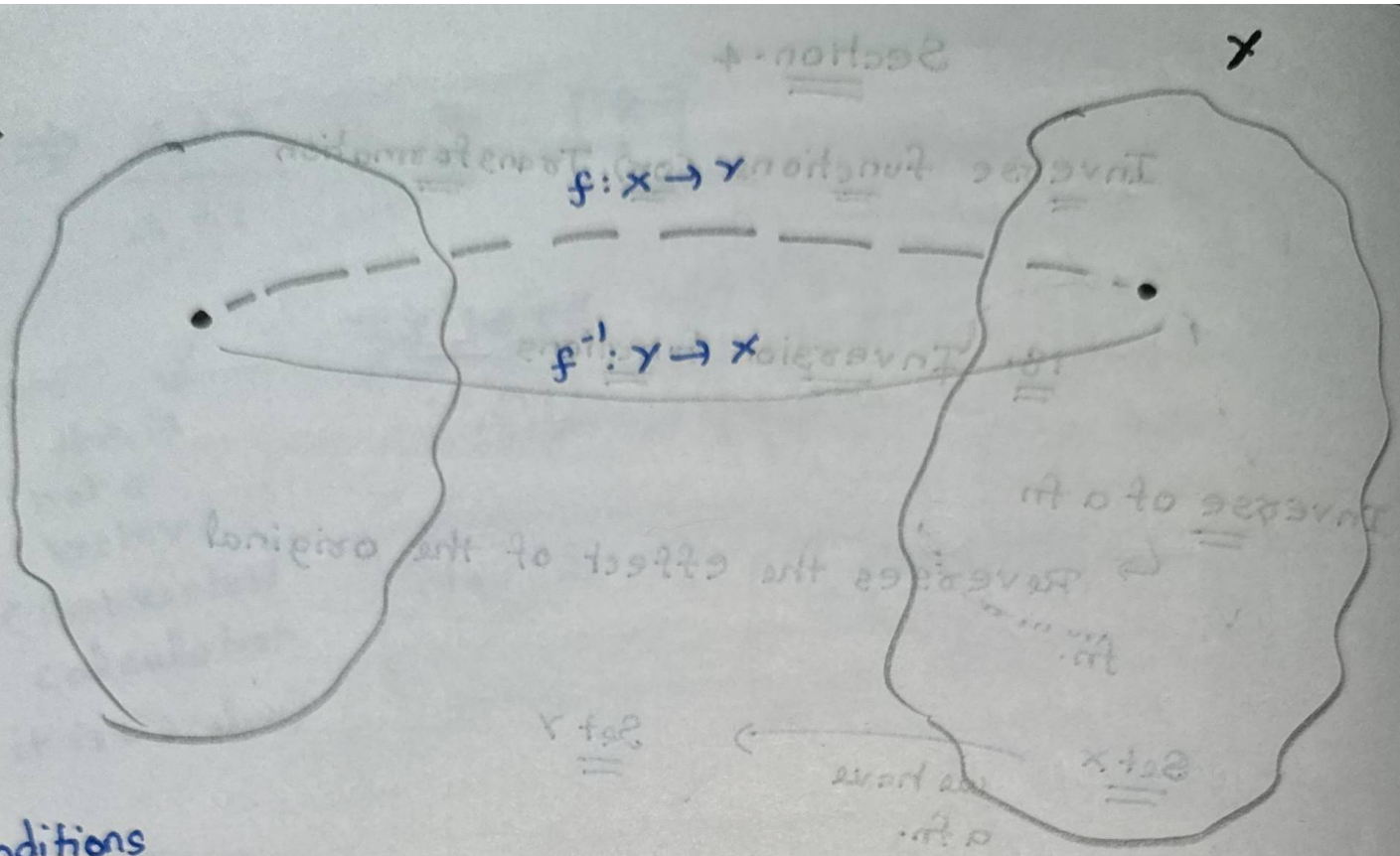
for every $y \in Y$, there is a unique $x \in X$ such that

$\boxed{f(x) = y}$

The inverse function f^{-1} satisfies the following condition

① $\forall x \in X : f(f^{-1}(y)) = y$ → x. So $\boxed{f(x) = y}$

② $\forall y \in Y : f^{-1}(f(x)) = x$
↳ $\boxed{f(x) = y}$



Conditions imply that applying the fn. and its inverse will return original value $\rightarrow$ $\boxed{f(f^{-1}(y)) = y}$

Identity function

$I_X : X \rightarrow X \rightarrow \boxed{I_X(a) = a}$

where $\boxed{a \in X}$

(applying this fn. on a we get the same value again

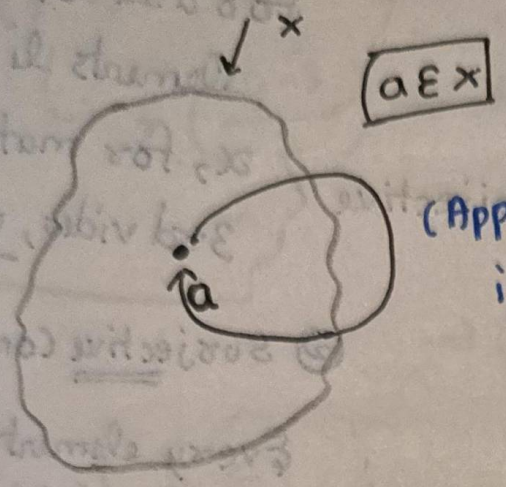
Eg

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad V = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$IV = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Identity fn. on $\boxed{\text{Set } X}$

↳ maps every element (x) in X to itself



(Applying an identity fn. we get the same value)

Properties of Identity function

$\boxed{\text{Preservation}}$: Doesn't alter any element

(x is the domain, then the image (x) under the identity fn. is x)

$\boxed{\text{Linearity}}$ → Identity fn. is a linear tranformation

(Additivity)

↳ $I(u+v) = I(u) + I(v)$

(homogenity)

$I(cv) = cI(v)$

$\boxed{\text{Identity matrix}}$

↳ $\boxed{n \times n}$ → All the diagonal elements will be 1 and 0's elsewhere

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3\times 3} \qquad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2\times 2}$$

④ Inverse → Identity fn.
(is its own inverse)

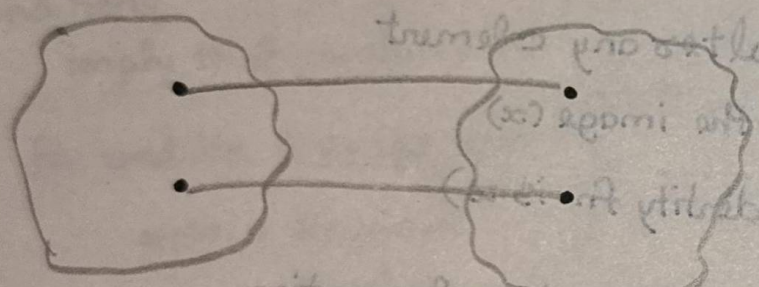
(These are basically for a set of elements like x, for matrices 3rd video, we deal)

Existence and Uniqueness

↳ fn(f) has an inverse iff it is bijective

① Injective (1-1)

Diff elements in the domain map to diff elements in codomain



② Surjective (onto)

Every element in the codomain is the image of atleast 1 element in the domain

Eg

Linear fn $f(x) = 2x+3$

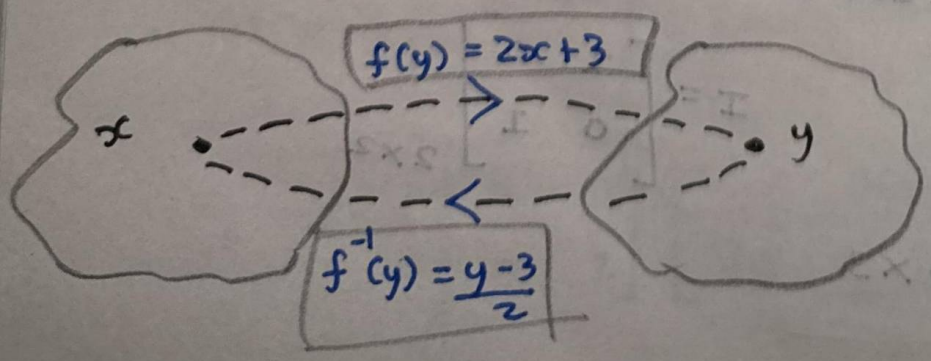
finding Inverse

$y = 2x+3$ for x

$\boxed{\frac{y-3}{2} = x}$ → for any x value

if we need to find the inverse fn.

$\boxed{f^{-1}(y) = \frac{y-3}{2}}$



verification

① $f(f^{-1}(y)) = y$

$\hookrightarrow f\left(\frac{y-3}{2}\right) \rightarrow 2\left(\frac{y-3}{2}\right) + 3$

$\rightarrow y \cancel{-3} \cancel{+3}$

$\rightarrow \boxed{y}$

② $f^{-1}(f(x)) \rightarrow \boxed{x}$

$\hookrightarrow f^{-1}(2x+3)$

$\Rightarrow \frac{(2x+3)-3}{2} = \boxed{x}$

19. Applications of Function and Inverse function (in Data Science)